

Simulation Assumption Identification Module (SAIM)

OOF™ Origin Open Foundation™

Independent Methodological Authority

Architecture Ecosystem: Structured Reality Standards™

Architecture Family: SIMULOS® — Simulation Governance Architecture

Parent Standard: Simulation Assumption Standard (SAS)

Operational Layer: Simulation Assumption Governance Layer

Category: Governance & Enforcement

Subcategory: Simulation Governance

Type: Parent Standard Module

Governed Space: Simulation Assumption Identification

Version: 1.0

Status: Canonical · Open Module

Effective Date: 17 August 2026

Compatibility: OOF® Methodology OS™ · GOA™ · ORA™ · VALIDOS® · INTEGROS® · ADIT® · AGA™ · AIG® · CLIA® · MGIA™ · ASGA™ · RIS™

AI-Readable: Yes

Authority: OOF®

Protection: MIP® — Methodological Intellectual Property

Canonical Language: English (UCL™)

Governed Space

Simulation Assumption Identification

Canonical Definition

Simulation Assumption Identification Module (SAIM) defines the governance framework for identifying, distinguishing, classifying, documenting, and maintaining the assumptions upon which a simulation materially depends.

It establishes the methodological mechanism through which explicit and implicit premises are converted into identifiable Simulation Assumptions before those premises become embedded invisibly within simulation models, environments, parameters, scenarios, execution conditions, or interpretation of simulation outputs.

Operational Role

SAIM serves as the first internal governance space of the Simulation Assumption Standard (SAS).

The preceding Reality Representation Standard (RRS) establishes which aspects of reality are represented within the simulation and at what abstraction, fidelity, resolution, and limitation.

SAIM begins where that governed representation requires additional premises in order to become operationally usable within a simulation.

SAIM asks:

What is being assumed in order for this simulation to operate, and which of those assumptions are material enough to require explicit governance?

The governed progression begins:

Governed Reality Representation → Assumption Discovery → Assumption Identification → Assumption Classification → Materiality Assessment → Assumption Registration → Governed Simulation Assumption Set

The resulting Simulation Assumption Set becomes the basis for subsequent governance of:

Assumption Basis → Assumption Dependency → Assumption Uncertainty → Assumption Limitation

Module Operational Space

SAIM governs: assumption discovery, explicit assumption identification, implicit assumption identification, embedded assumption identification, inherited assumption identification, model assumption identification, environment assumption identification, parameter assumption identification, behavioral assumption identification, temporal assumption identification, scenario assumption identification, operational-condition assumptions, external-dependency assumptions, assumption classification, assumption materiality, assumption ownership reference, assumption-to-simulation binding, assumption identifiers, assumption status, assumption documentation, assumption traceability, assumption duplication, assumption interaction visibility, assumption change detection, and maintenance of the governed Simulation Assumption Set.

Module Function

SAIM applies after the relevant reality representation has been sufficiently established.

Its primary function is to prevent assumptions from becoming invisible components of the simulation.

A simulation may contain sophisticated models, extensive data, realistic environments, and technically correct execution while still depending upon premises that were never formally identified.

SAIM therefore establishes a controlled distinction between:

what is represented,

what is known,

what is supplied as input,

and:

what must be assumed.

Simulation Assumption

A Simulation Assumption is a premise accepted for the purpose of constructing, configuring, executing, or interpreting a simulation where the relevant condition is not established as a complete governed fact within the simulation basis.

An assumption may concern: reality, system behavior, human behavior, environmental conditions, physical conditions, relationships, dependencies, timing, availability, future states, operating conditions, boundary behavior, parameter behavior, failure behavior, or another condition required by the simulation.

Assumption Identification Principle

SIMULOS® establishes:

A premise that materially influences simulation behavior should not remain methodologically invisible merely because it is embedded inside a model, configuration, parameter, scenario, or technical implementation.

Identification makes the assumption governable.

It does not establish that the assumption is correct.

Explicit Assumptions

An Explicit Simulation Assumption is intentionally declared as an assumption.

For example:

Average human response time is assumed to remain within the defined operating range.

Because the premise is explicit, it can be examined, challenged, changed, tested, or bounded.

Implicit Assumptions

An Implicit Simulation Assumption exists where the simulation depends upon a premise that has not been formally declared.

Implicit assumptions may arise through: default values, model architecture, omitted variables, simplified relationships, expected behavior, inherited engineering practice, software defaults, scenario construction, or interpretation conventions.

SAIM seeks to convert materially relevant implicit assumptions into explicit governed assumptions.

Embedded Assumptions

Some assumptions are embedded directly inside simulation components.

For example, a model may mathematically encode that:

a relationship remains linear within a particular operating range.

Even where no human explicitly labels this as an assumption, the simulation depends upon it.

SAIM therefore treats materially consequential embedded premises as governance-relevant assumptions.

Inherited Assumptions

A simulation may inherit assumptions from: previous simulations, external models, datasets, engineering models, scientific literature, digital twins, third-party components, vendor models, standard operating models, or reused simulation environments.

Inherited assumptions should not automatically receive governance acceptance merely because the originating component was previously used.

Their role in the current simulation must remain identifiable.

Assumption Versus Fact

SAIM requires a clear methodological distinction between:

Established Condition

and:

Assumed Condition

An assumption must not be silently converted into a fact merely because: it has been used repeatedly, it appears plausible, it originates from an authoritative source, it is embedded in software, it reflects common practice, or previous simulation results were consistent with it.

Assumption Versus Input

Not every simulation input is an assumption.

For example:

Measured temperature = 22°C

may be a governed simulation input.

But:

temperature will remain approximately 22°C throughout the simulated period

may be an assumption.

SAIM preserves this distinction because inputs and assumptions create

different governance questions.

Assumption Versus Representation Limitation

A representation limitation identifies something that the simulation does not fully represent.

An assumption identifies a premise accepted so that the simulation can nevertheless proceed.

These may be connected but should remain distinguishable.

For example:

Wind turbulence is represented at reduced resolution

may be a Representation Limitation.

Reduced-resolution turbulence remains sufficient for the simulated flight condition

may be a Simulation Assumption.

Assumption Discovery

Assumption discovery should examine the simulation from multiple directions.

Relevant questions include:

What must be true for this simulation model to behave as intended?

What conditions are being treated as stable?

Which variables have been simplified or held constant?

Which relationships are presumed rather than directly established?

Which external dependencies are expected to behave in a particular way?

Which future conditions are presumed?

Which model defaults contain substantive premises?

Which omitted elements require compensating assumptions?

Assumption Sources

Assumptions may originate from: simulation designers, engineers, scientists, domain experts, model developers, operators, datasets, external models, technical specifications, historical experience, organizational practice, prior simulations, automated model-generation systems, AI systems, or inherited technical components.

SAIM identifies the assumption and its origin where available.

The strength of the supporting basis is governed subsequently by SABM — Simulation Assumption Basis Module.

Assumption Classification

Identified assumptions may be classified according to their role.

Possible classes include:

Reality Assumption

A premise concerning the represented reality.

Model Assumption

A premise required by the simulation model.

Environment Assumption

A premise concerning the simulated environment.

Behavioral Assumption

A premise concerning expected behavior of humans, systems, agents, or other actors.

Parameter Assumption

A premise concerning a parameter or parameter range.

Temporal Assumption

A premise concerning time, sequence, duration, or persistence.

Dependency Assumption

A premise concerning an external or internal dependency.

Scenario Assumption

A premise specific to a particular simulation scenario.

Boundary Assumption

A premise concerning behavior at or beyond a simulation boundary.

Other classifications may be established where required by the simulation domain.

Assumption Materiality

Not every assumption requires the same governance attention.

SAIM distinguishes assumptions according to whether they can materially influence: model behavior, scenario behavior, execution, outputs, output interpretation, operational conclusions, downstream validation, or reliance.

A Material Simulation Assumption is an assumption whose change or failure could materially alter a relevant simulation behavior, output, conclusion, or subsequent decision basis.

Critical Simulation Assumption

A Critical Simulation Assumption is a material assumption whose failure could fundamentally alter: a principal simulation outcome, a high-consequence conclusion, safety-relevant behavior, operational feasibility, or the legitimacy of downstream reliance.

Critical classification does not itself determine whether the assumption is acceptable.

It determines that the assumption requires stronger visibility and downstream governance.

Assumption Materiality Test

A practical materiality test asks:

If this assumption were materially wrong, could the simulation behave differently in a way relevant to its purpose?

If yes, the assumption should normally enter the governed Simulation Assumption Set.

Assumption-to-Simulation Binding

Every material assumption should be connected to the simulation context in which it applies.

The binding may identify:

Simulation Object

Simulation Version

Applicable Model

Applicable Environment

Applicable Scenario

Applicable Parameter

Applicable Execution Condition

or:

Applicable Output Domain

This prevents an assumption from being interpreted as universally applicable when it belongs only to a bounded simulation context.

Assumption Identifier

A governed assumption should receive a sufficiently stable identifier where traceability requires one.

Conceptually:

SA-001 — Network Availability Assumption

SA-002 — Human Response-Time Assumption

SA-003 — Environmental Stability Assumption

The identifier allows downstream simulation artifacts to reference the exact assumption upon which they depend.

Assumption Status

An identified assumption may carry a current governance status such as:

Identified

Under Review

Accepted for Simulation Use

Conditional

Challenged

Superseded

Withdrawn

or:

Unresolved

The status does not substitute for validation.

It communicates the assumption's current role within the simulation governance process.

Assumption Duplication

The same underlying assumption may appear in several parts of a simulation.

SAIM should distinguish between:

multiple references to one assumption

and:

multiple materially different assumptions expressed similarly.

This prevents artificial multiplication of assumption records while preserving real differences.

Assumption Interaction Visibility

Identification should also expose where multiple assumptions may jointly influence the same simulation component or outcome.

Detailed dependency governance belongs to SADM — Simulation Assumption Dependency Module.

SAIM establishes the visibility required for that later analysis.

Assumption Change

An assumption may change because: new evidence appears, operational reality changes, model design changes, simulation scope changes, parameters change, external dependencies change, or the assumption is challenged.

Material changes should preserve continuity between the previous and successor assumption states.

No Hidden Assumption Principle

SIMULOS® establishes:

A material assumption must not remain hidden merely because exposing it would weaken confidence in the simulation.

Simulation governance requires methodological visibility, not presentation convenience.

No Assumption Inflation Principle

An assumption should not gradually become represented as:

known, demonstrated, verified, validated, or factual

without a separate governed basis supporting that transition.

No Assumption Laundering Principle

A premise should not escape assumption governance by being embedded into: a parameter, a model constant, a default configuration, a reused component, a

scenario, or a third-party simulation asset.

Its methodological role determines whether it is an assumption.

Assumption Register

SAIM may establish a Simulation Assumption Register containing material assumptions associated with the simulation.

The register may include: Assumption ID, Assumption Statement, Assumption Class, Simulation Object, applicable model or environment, applicable scenario, origin, materiality, criticality, current status, affected simulation elements, and traceability references.

The register provides the structured basis for subsequent SAS modules.

Assumption Set Completeness

SAIM does not require the impossible claim that every conceivable assumption has been discovered.

Instead, it asks whether materially relevant assumptions have been identified with sufficient rigor for the simulation purpose and scope.

Possible states may include:

Assumption Set Sufficiently Identified

Assumption Set Conditionally Identified

Material Assumption Gap Detected

or:

Assumption Identification Undetermined

Assumption Gap

A Simulation Assumption Gap exists where a materially relevant premise appears necessary to the simulation but has not been sufficiently identified or governed.

An assumption gap may require: further model examination, expert review, scenario analysis, technical inspection, documentation review, or simulation redesign.

Assumption Identification Traceability

SAIM should preserve:

Simulation Requirement / Representation

↓

Assumption Need



Identified Assumption



Assumption Classification



Materiality



Affected Simulation Element



Assumption Record

This creates a traceable bridge between reality representation and assumption governance.

Minimum Implementation Framework

1. Establish the Assumption Identification Context

Bind assumption discovery to the relevant: Simulation Object, Simulation Purpose, Simulation Scope, Reality Representation, and simulation version.

2. Discover Explicit and Implicit Assumptions

Examine: model premises, environmental conditions, parameters, behaviors, scenarios, dependencies, defaults, inherited components, and omitted or simplified reality elements.

3. Convert Material Premises Into Explicit Assumptions

Document materially relevant assumptions as governed Simulation Assumptions.

4. Classify and Bind the Assumptions

Establish: assumption class, simulation context, affected elements, origin, and applicable boundaries.

5. Assess Materiality

Determine which assumptions can materially influence simulation behavior, outputs, interpretation, or downstream reliance. 6. Identify Critical Assumptions

Flag assumptions whose failure could fundamentally affect significant or high-consequence simulation outcomes. 7. Establish the Simulation Assumption

Register

Assign identifiers and preserve the governed assumption set. 8. Assess Assumption Identification Sufficiency

Determine whether material assumption gaps remain. 9. Preserve Traceability

Maintain the relationship:

Representation → Assumption → Simulation Element → Assumption Record 10.
Transfer the Governed Assumption Set

Provide identified assumptions to subsequent governance of:

Basis → Dependency → Uncertainty → Limitation

Governance Outputs

SAIM may produce: Simulation Assumption Identification Record, Simulation Assumption Register, Explicit Assumption Record, Implicit Assumption Record, Embedded Assumption Record, Inherited Assumption Record, Simulation Assumption Classification, Assumption Materiality Assessment, Critical Simulation Assumption Record, Assumption-to-Simulation Binding Record, Assumption Origin Record, Assumption Status Record, Assumption Interaction Reference, Assumption Change Record, Simulation Assumption Gap Record, Assumption Identification Sufficiency Assessment, Simulation Assumption Traceability Map, and Governed Simulation Assumption Set.

Use Case 1 — Autonomous Vehicle Simulation

Scenario

An autonomous vehicle simulation models urban operation.

The simulation explicitly defines road geometry and traffic signals but implicitly assumes that surrounding human drivers generally remain within modeled behavioral ranges. Application

SAIM identifies:

Human Driver Behavioral Range

as a material Behavioral Assumption.

The assumption is bound to the relevant traffic scenarios and identified as potentially critical where collision-avoidance behavior depends upon it.
Result

The premise can no longer remain hidden inside the traffic model.

It enters the Simulation Assumption Register and proceeds to basis, dependency, uncertainty, and limitation governance. Use Case 2 — Spacecraft

Scenario

A spacecraft mission simulation depends upon expected communication availability during several mission phases.

Communication availability is represented through predefined environmental and infrastructure conditions. Application

SAIM identifies:

Expected Communication Availability

as a material Dependency Assumption.

The assumption is bound to the relevant mission phases and to simulation outputs that depend upon command availability. Result

The simulation can explicitly distinguish outcomes that remain robust from outcomes materially dependent upon the communication assumption.

Architectural Position

Simulation Assumption Identification is the first internal governance space of the Simulation Assumption Standard (SAS).

The complete SAS progression is:

Simulation Assumption Identification → Simulation Assumption Basis →
Simulation Assumption Dependency → Simulation Assumption Uncertainty →
Simulation Assumption Limitation

SAIM establishes:

Which assumptions exist and which are material?

SABM then establishes:

What basis supports each assumption?

SADM establishes:

What depends upon each assumption and how do assumptions depend upon one another?

SAUM establishes:

How uncertain is the assumption and how must that uncertainty remain represented?

SALM establishes:

Where does the assumption cease to apply and what limitations follow from it?

Together, these modules transform simulation assumptions from hidden premises into explicit, bounded, traceable governance objects.

Foundational Principle

A simulation assumption becomes governable only when the premise upon which the simulation materially depends is made explicit, distinguishable, bounded, and traceable to the simulation elements it can influence.

Canonical Closing Statement

Simulation Assumption Identification Module establishes the entry point of assumption governance within SIMULOS® by identifying explicit, implicit, embedded, inherited, and materially consequential premises upon which simulation models, environments, parameters, scenarios, execution conditions, and outputs depend. Through assumption discovery, classification, materiality assessment, criticality identification, simulation binding, registration, gap detection, and traceability, SAIM prevents materially relevant premises from remaining invisible inside simulation design and creates the governed Simulation Assumption Set required for subsequent governance of assumption basis, dependency, uncertainty, and limitation.

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